MIL-F-16377/61(SH)
14 May 1982
SUPERSEDING
MS17338(SHIPS) in part
29 September 1967
(See page 8)

MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, FLOODLIGHTS

300 WATTS, 120 VOLTS, DRIPPROOF 15 DEGREES, SYMBOL 303.1

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH)

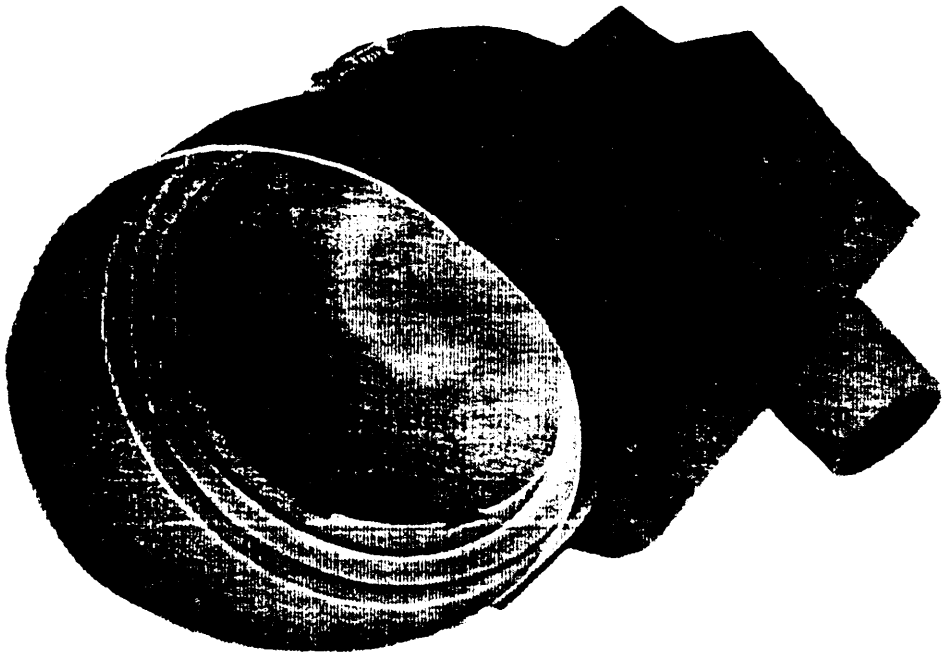


FIGURE 1. Typical view of Symbol 303.1.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 623

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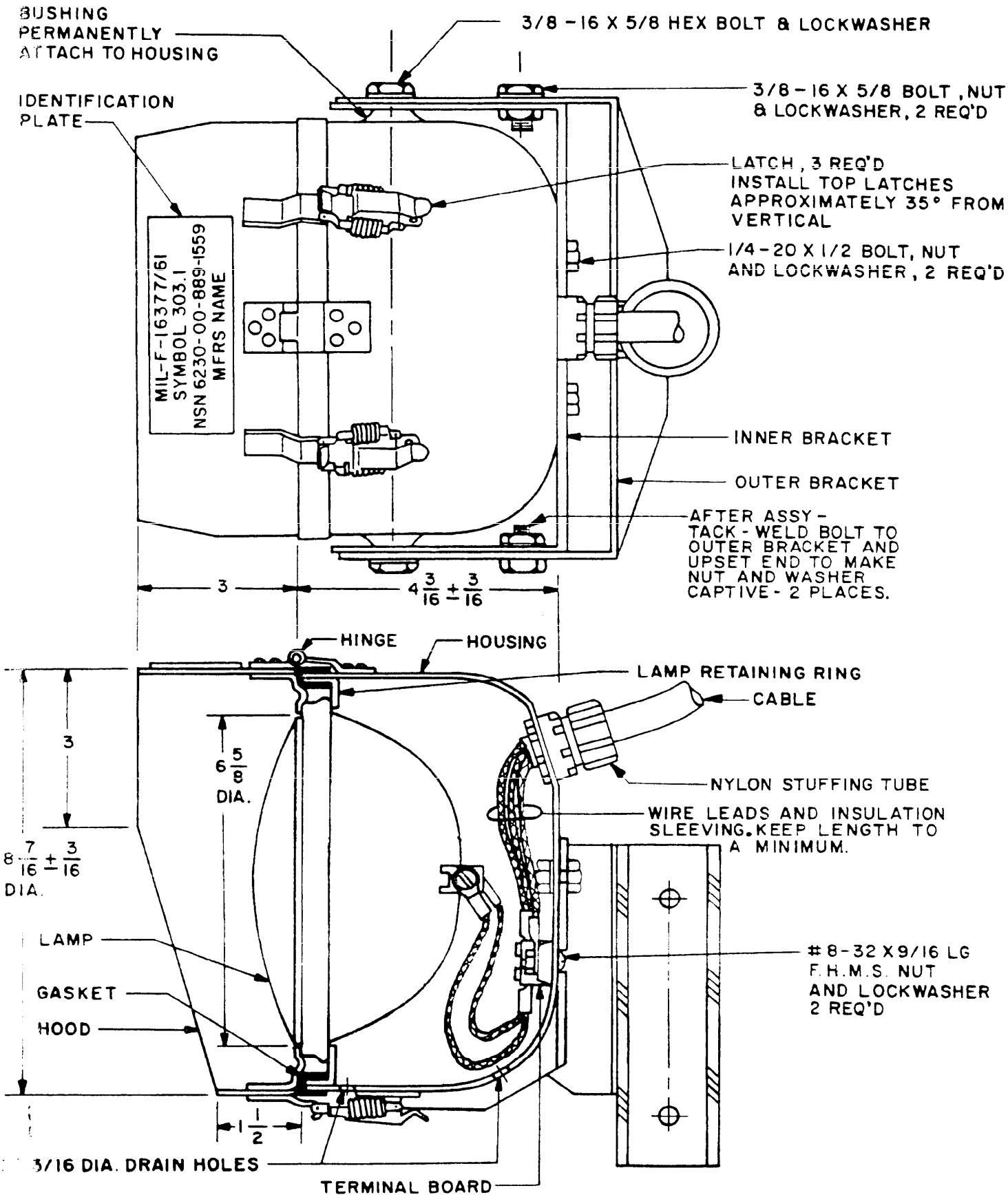


FIGURE 2. Typical top and section view.

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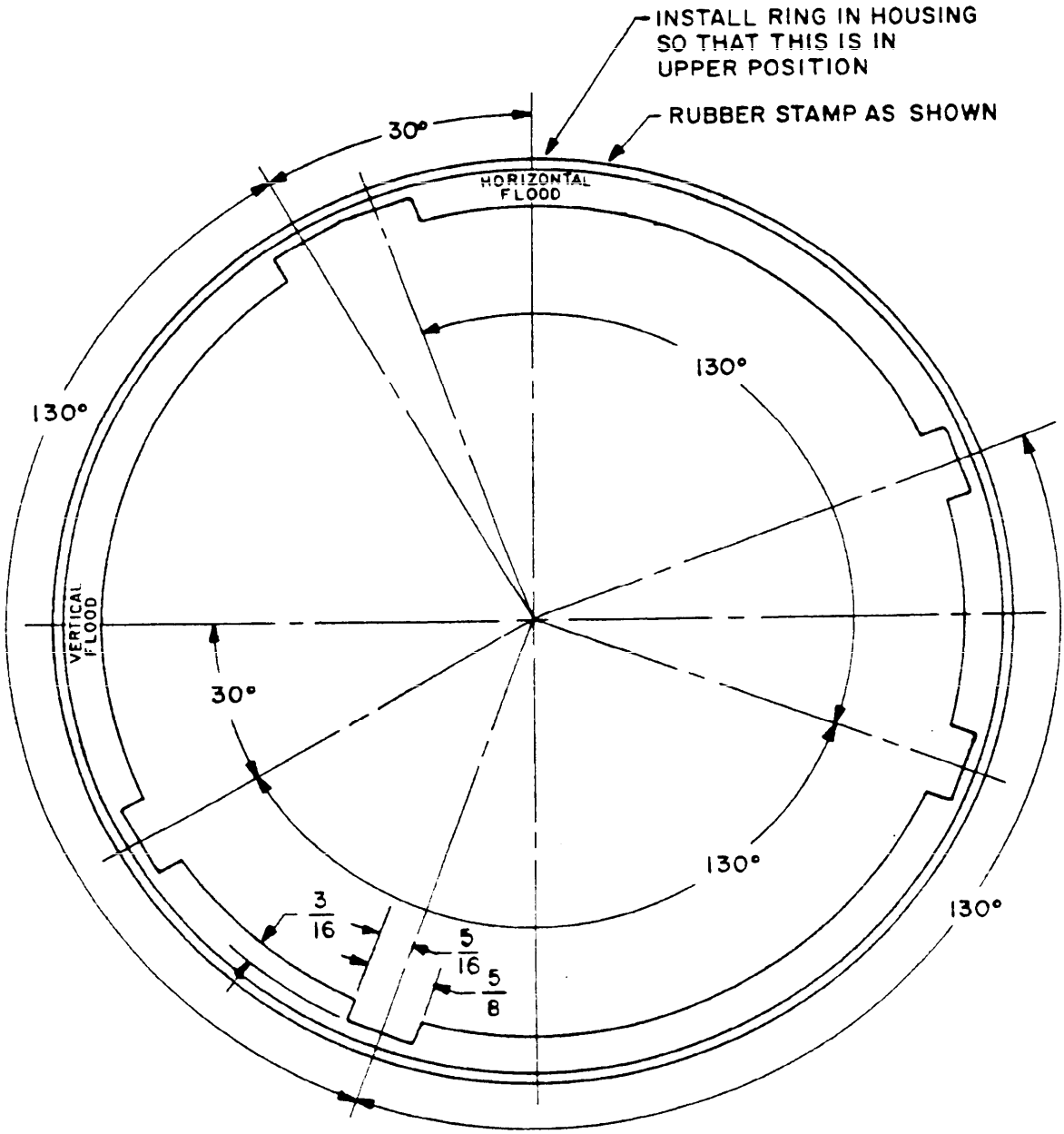


FIGURE 3. Lamp retaining ring.

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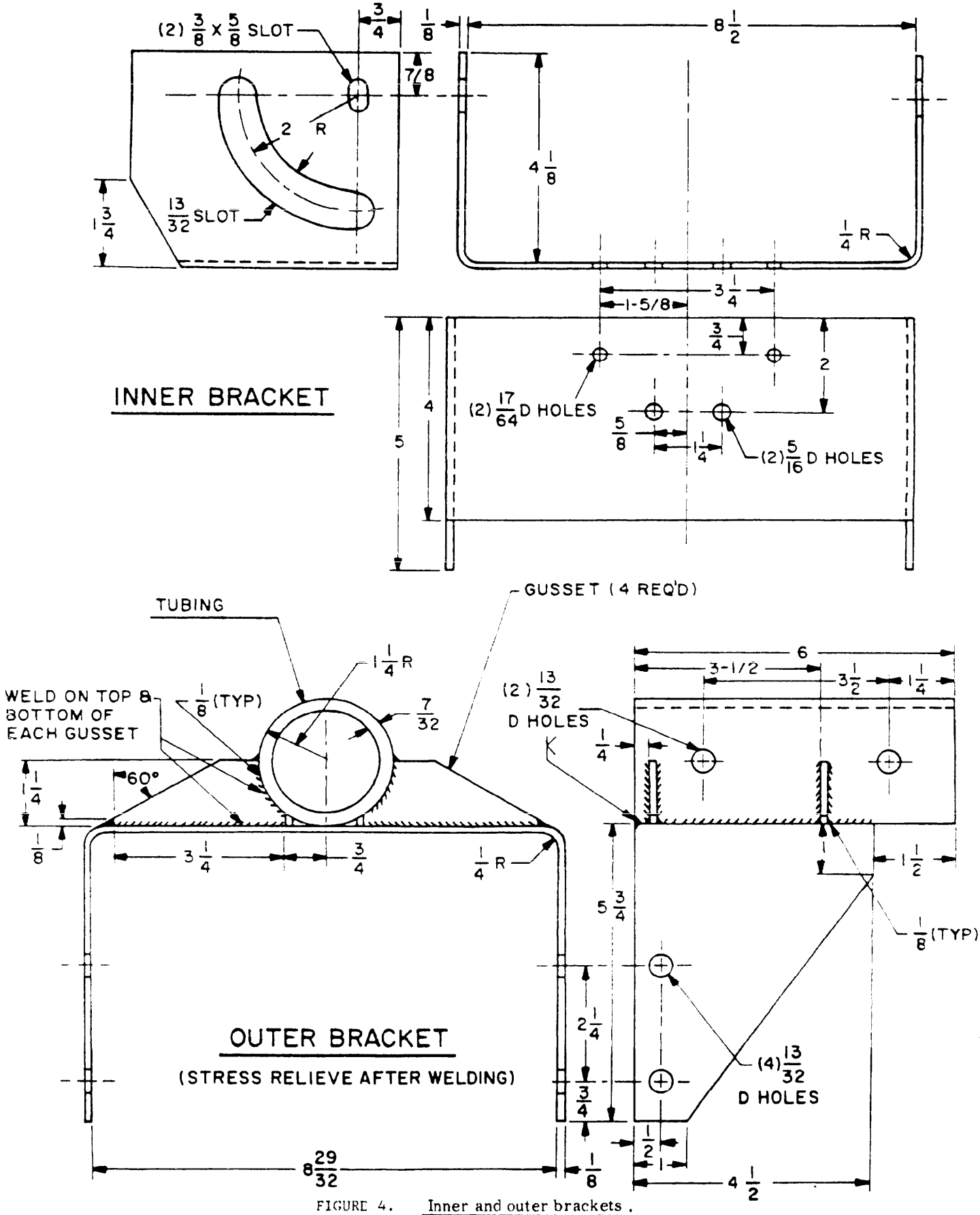


FIGURE 4. Inner and outer brackets.

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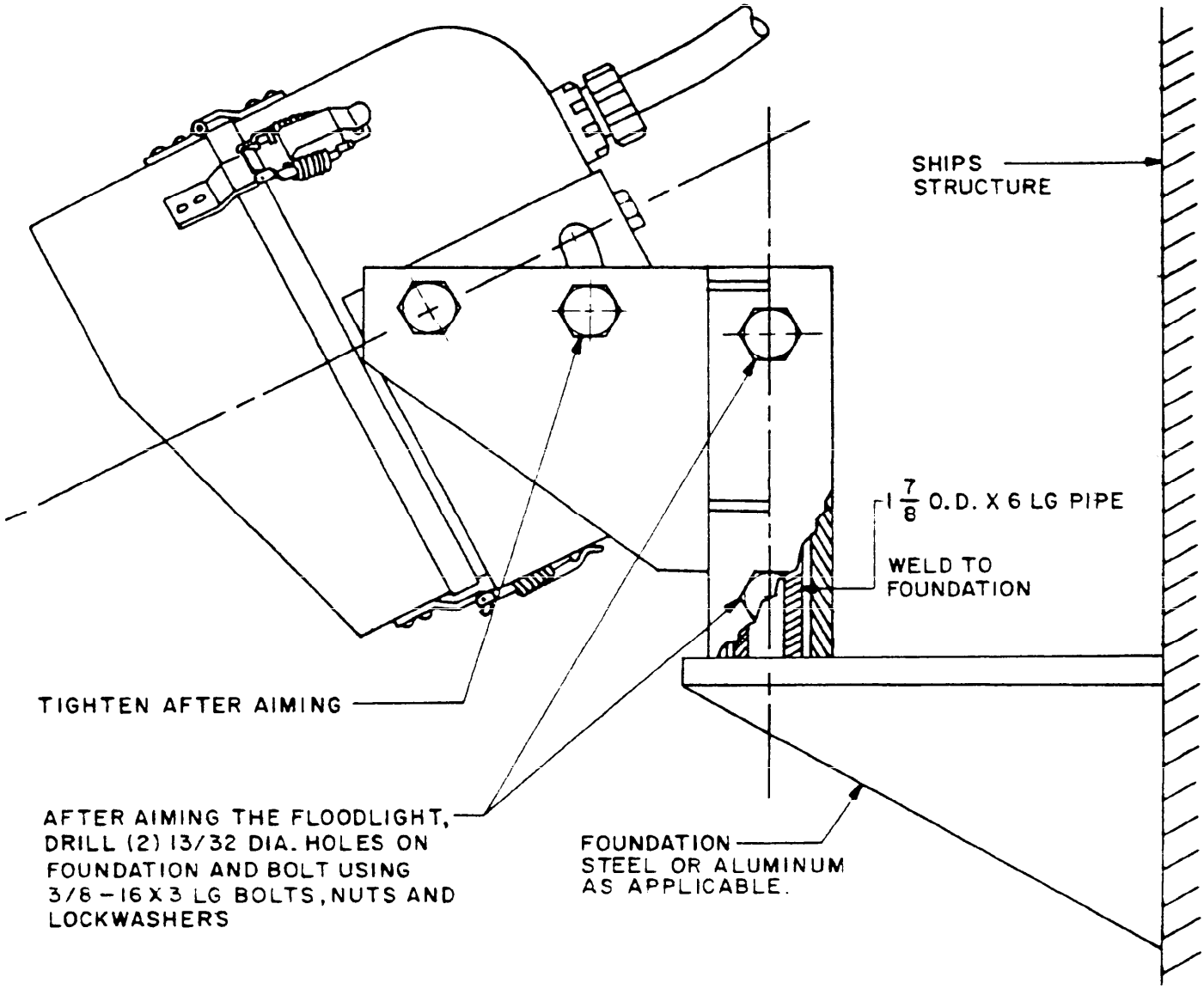


FIGURE 5. Typical mounting detail.

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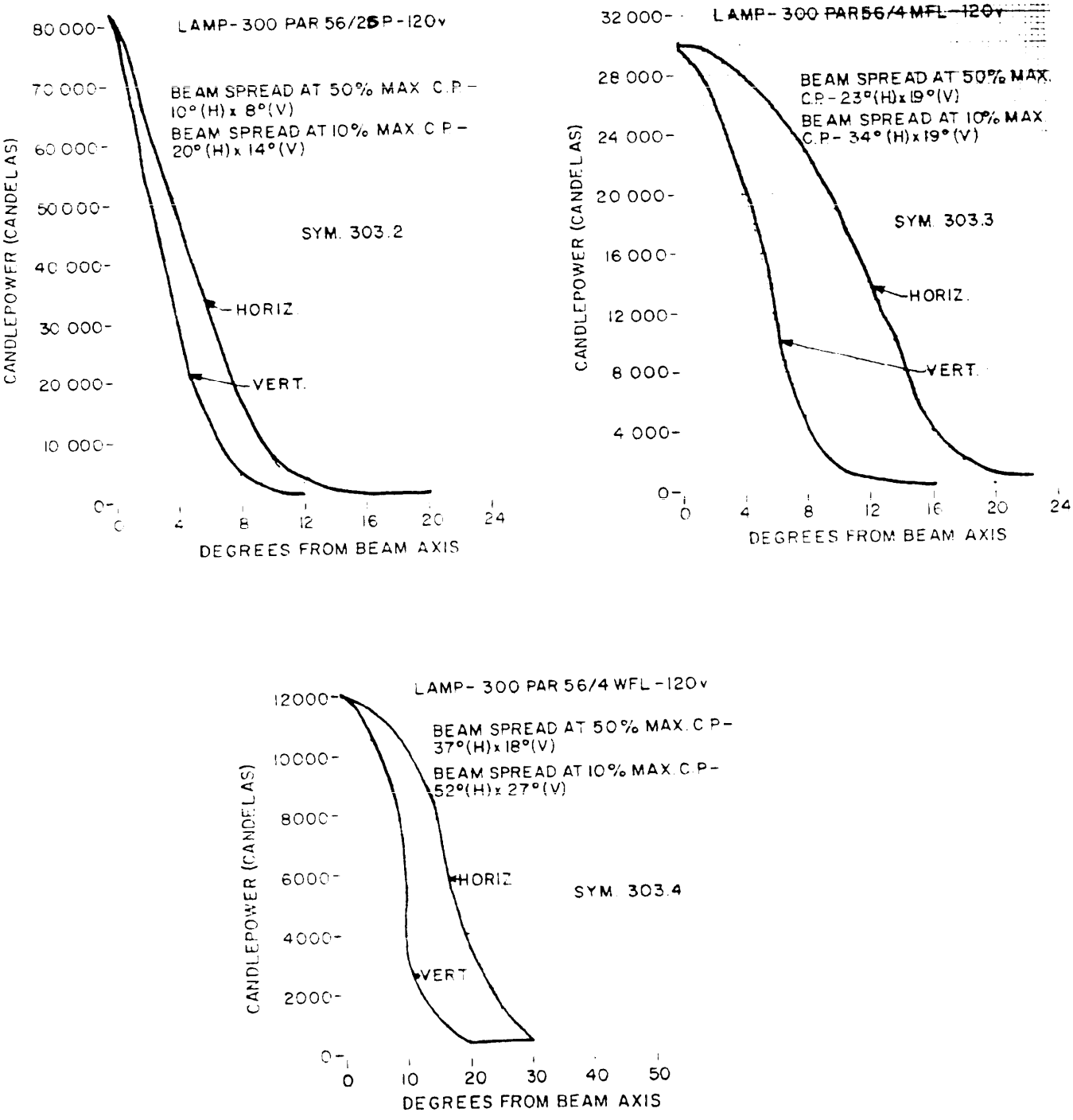


FIGURE 6. Typical candlepower distribution.

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REQUIREMENTS:

CLASSIFICATION:
Type: II
Class: 2

SYMBOL NUMBER:
303.1: Floodlight enclosure for PAR56 sealed beam lamps, white.

DESIGN:
Electrical rating: 300 watts, 120 volts.
Dimensions and configuration: See figures 1 through 6. Housing dimensions may be considered to be variable to the extent that available commercial tooling can be used.
Material: Corrosion resistant steel (CC 304) except as otherwise indicated in this specification. Material for lamp retaining ring may be phosphor bronze.
Enclosure: Drip-proof 15 degrees.
Lamp: See general information. Lamp shall not be furnished.
Cable: Floodlights shall be furnished with 10 feet of 3-conductor cable, type THOF-3 in accordance with MIL-C-915/6. The third conductor shall be used for grounding. The length of the wire leads t the terminal board and ground shall be kept to a minimum. Terminal lugs, MS17143-11, shall be usec
Insulation sleeving: Entire length of the cable wire leads to the terminal board and internal wiring shall be fitted with insulation sleeving conforming to MIL-I-003190/6.
Internal wiring: The enclosure shall be furnished with two wire leads, 15 inches long, attached to th terminal board. These leads are for connection to the lamp, and both ends shall be fitted with terminal lugs conforming to MS17143-11. The wire leads shall be #16 AWG, stranded, 250°C application, nickel clad copper, teflon jacket with overbraid of silicone glass braid dip.
Terminal board: The terminal board shall be type 8TB2HT of Military Specification Sheet MIL-T-55164/1
Nylon stuffing tube: Nylon stuffing tube shall be Size 2, part number M19622/1-002 of Military Specification Sheet MIL-S-19622/1. Packing assembly and "O" ring shall be as specified in MIL-S-19622/1 for Size 2 tube.
Weight: 14 pounds maximum
Designation and marking: An identification plate shall be furnished in accordance with MIL-F-16377. Information and location shall be as shown on figure 2.
Finish: Gray enamel

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:
First article inspection:
Complete floodlights: See table I.
Quality conformance:
Comparison inspection: Same as first article inspection.
Inspection of product for delivery.
Examination: See MIL-F-16377.
Examination for preparation for delivery: See MIL-F-16377.
Quality conformance tests: See MIL-F-16377.

GENERAL INFORMATION:
(a) Assignment of Military part numbers and general information is shown in table II.
(b) Supersession data. MS17338 has been superseded as follows:
Symbol 300.2 of MS17338 superseded by M16377/61-303.1.
Symbol 303 of MS17338 superseded by Symbol 263 of Naval Air Engineering Center drawing NAEC(SI) 506829-1.
(c) This floodlight is so designed that a typical lamp, having a beam spread of 60° horizontal, 30° vertical can be repositioned in the floodlight to obtain a reverse beam spread of 30° horizontal 60° vertical by rotating the lamp 90° in either direction.

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TABLE I. First article and comparison inspection for complete floodlights.

Tests	Requirement paragraph of MIL-F-16377	Test paragraph of MIL-F-16377	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14	1/ DP-15 degrees
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14	1/ DP-15 degrees
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Grounding circuit	3.6.13.3	4.8.11	
Salt spray	3.5.15	4.8.10	

1/ Floodlight for this test shall be mounted in a manner similar to that shown on figure 5.

TABLE II. Assignment of part numbers and general information.

Military Part Number M16377/61-	Item name	National Stock Number
303.1	Complete floodlight as detailed in this specification but without lamp, Symbol 303.1.	6230-00-889-1559
303.2	Symbol 303.1 with 300w, 120v spot lamp, Industry Number 300PAR56/2SP, MS15535-4; Beam spread: 20° horizontal, 15° vertical.	6230-00-889-1559 6240-00-270-4689
303.3	Symbol 303.1 with 300w, 120v medium flood lamp, Industry Number 300PAR56/4MFL; Beam spread: 35° horizontal, 26° vertical.	6230-00-889-1559 6240-00-145-1148
303.4	Symbol 303.1 with 300w, 120v wide flood lamp, Industry Number 300PAR56/4 WFL, MS15535-6; Beam spread: 60° horizontal, 30° vertical.	6230-00-889-1559 6240-00-056-0737
303.5	Symbol 303.1 with 250w, 24-28v medium flood lamp, Industry Number 4549, MS15535-3; Beam spread: 40° horizontal, 18° vertical.	6230-00-889-1559 6240-00-270-4695

PREPARING ACTIVITY:
NAVY-SH
(PROJECT 6230-N269)

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